

STEFFANY NERY

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EDUCATION

Lamar University | Bachelor of Science, Chemical Engineering

Beaumont, TX

GPA: 3.85 / 4.00

Expected May 2029

Relevant Coursework: Introduction to Chemical Engineering (distillation towers, material balances), Chemical Engineering Applied Math (Excel, VBA, Polymath, linear algebra, regression, Aspen Plus steady state simulation; in progress), Engineering Economics, Calculus III, Chemistry I and II, Physics I and II, Organic Chemistry

High School: Sabine Pass ISD, Valedictorian, Class of 2025

ENGINEERING PROJECTS

Multicomponent Distillation Column Simulation

Summer 2026

Independent project | DWSIM 9.0.5, Peng Robinson equation of state

- Built and converged a rigorous 20 stage distillation column separating a five component C3 to C7 light hydrocarbon feed at 5,000 kmol/h
- Defined light and heavy key components (isopentane / n hexane), product purity specifications, and a reflux ratio of 1.5; solved for condenser and reboiler duties (39.4 MW and 44.9 MW), stage temperature and pressure profiles, and product compositions
- Met both product purity targets: n hexane impurity in distillate at 0.08 mol% and isopentane impurity in bottoms at 0.25 mol%, both well below the 1 mol% specification; achieved 99.5% isopentane recovery and 99.9% n hexane recovery
- Authored a full technical report documenting methodology, results, and design recommendations; verified component mass balance closure across all five species to within 0.1%

EXPERIENCE

Engineering Assistant, Avocado Farm Project

June 2024 to July 2024

Private Agricultural Site, Peribán, Mexico

- Collaborated with an agricultural engineer to monitor tree health and identify diseased or stressed trees
- Collected and analyzed soil, bark, and environmental samples to diagnose plant health issues; supported solutions that improved crop productivity and nutrient balance

Peer Tutor, Physics, Calculus I and II, Chemistry, and Spanish

January 2022 to May 2025

Sabine Pass ISD, Sabine Pass, TX

- Tutored classmates one on one and in small groups across four STEM and language subjects over three years

TECHNICAL SKILLS

Process Simulation: DWSIM (rigorous distillation, thermodynamic property packages); Aspen Plus (in progress, fall 2026)

Data Analysis: Microsoft Excel, Polymath, engineering plotting and data presentation

Certification: Microsoft Office Specialist: Word 2016 (Microsoft, December 2022)

Software: Microsoft Office Suite (Word, Excel, PowerPoint), Adobe Photoshop

Languages: Bilingual, English and Spanish

HONORS AND ACTIVITIES

- Recipient of five academic scholarships: Lamar Academic Excellence, Women in Industry, Rotary Club, Sabine Pass Education Foundation, and The University of Texas at Austin
- Society of Women Engineers (SWE), active member; American Institute of Chemical Engineers (AIChE), member
- Two time UIL State Champion, News Writing; 1st Place Region, UIL Computer Science; team captain for Computer Science, News Writing, and Feature Writing
- National Honor Society; Citizenship Award and President's Award
- Volunteer: Lamar Engineering Student Picnic, beach cleanups and local environmental initiatives, nursing homes